How do Large Language Models Handle Multilingualism?

Yiran Zhao[1,2†] **Wenxuan Zhang**[2,3‡] **Guizhen Chen**[2,4§] **Kenji Kawaguchi**[1] **Lidong Bing**[2,3]
[1] National University of Singapore [2] DAMO Academy, Alibaba Group, Singapore
[3] Hupan Lab, 310023, Hangzhou, China [4] Nanyang Technological University, Singapore

Abstract

Large language models (LLMs) have demonstrated impressive capabilities across diverse languages. This study explores how LLMs handle multilingualism. Based on observed language ratio shifts among layers and the relationships between network structures and certain capabilities, we hypothesize the LLM's multilingual workflow (`MWork`): LLMs initially understand the query, converting multilingual inputs into English for task-solving. In the intermediate layers, they employ English for reasoning and incorporate multilingual knowledge with self-attention and feed-forward structures, respectively. In the final layers, LLMs generate responses aligned with the original language of the query. To verify `MWork`, we introduce Parallel Language-specific Neuron Detection (`PLND`) to identify activated neurons for inputs in different languages without any labeled data. Using `PLND`, we validate `MWork` through extensive experiments involving the deactivation of language-specific neurons across various layers and structures. Moreover, `MWork` allows fine-tuning of language-specific neurons with a small dataset, enhancing multilingual abilities in a specific language without compromising others. This approach results in an average improvement of 3.6% for high-resource languages and 2.3% for low-resource languages across all tasks with just 400 documents.[1]

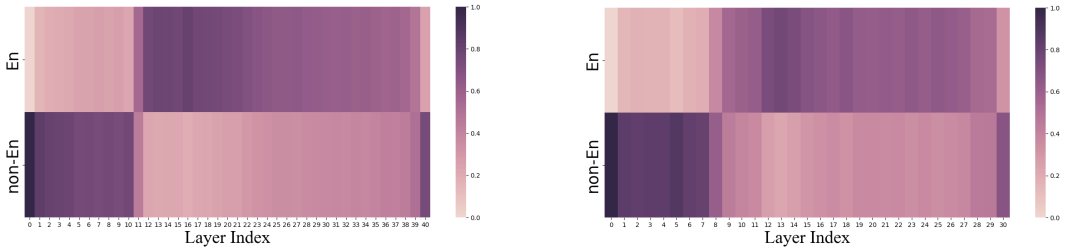
1 Introduction

Recent advancements in large language models (LLMs) (OpenAI, 2023; Touvron et al., 2023; Team et al., 2023) have dramatically transformed the field of natural language processing (NLP). Thanks to the extensive pretraining on massive corpora mixed with different languages, these models demonstrate remarkable capabilities in understanding and generating text across multiple languages (Huang et al., 2023; Zhang et al., 2023a; Zhao et al., 2024a). Despite these advancements, the intricate mechanism of their multilingual processing behavior remains largely unclear, which leads to an important research question: *How do large language models handle multilingualism?*

To understand the working mechanism of LLMs, existing studies mainly focus on the relationship between model architectures and certain capabilities, with some investigating reasoning abilities with self-attention layers (Hou et al., 2023; Stolfo et al., 2023; Friedman et al., 2023), and others interpreting feed-forward layers as key-value memories for storing factual knowledge (Geva et al., 2021; Dai et al., 2022; Meng et al., 2022). However, these works solely center on English and neglect the multilingual features of LLMs in their interpretations.

[†]This work was done during the internship of Yiran Zhao at Alibaba DAMO Academy.
[‡]Wenxuan Zhang is the corresponding author: `isakzhang@gmail.com`
[§]Guizhen Chen is under the Joint Ph.D. Program between DAMO Academy and NTU.
[1]Our code is available at `https://github.com/DAMO-NLP-SG/multilingual_analysis`

38th Conference on Neural Information Processing Systems (NeurIPS 2024).



(a) Vicuna-13b-v1.5 (b) BLOOMZ-7b1

Figure 1: Ratio of English and non-English tokens among layers given non-English queries.

To gain an initial understanding of the multilingual mechanism of LLMs, we test LLMs with various non-English queries and decode the hidden embeddings of each layer to tokens within the LLM's vocabulary. Subsequently, we classify these decoded tokens into either English or non-English, and analyze the ratio. Figure 1 illustrates the ratio of English and non-English tokens for each layer of two LLMs. We observe that non-English queries initially generate non-English embeddings as expected. However, as queries progress through the middle layers, the representations surprisingly become English-centric. In the final layers, there is a reversion to predominantly non-English embeddings, matching the non-English queries.

Motivated by the observed transformation above, we hypothesize a three-stage multilingual workflow: *understanding*, *task-solving*, and *generating*. This involves understanding the original non-English queries and interpreting them in English, solving tasks in English, and reverting outputs back to the original language. Furthermore, building upon previous studies that link self-attention structures to reasoning and feed-forward structures to factual knowledge storage (Hou et al., 2023; Geva et al., 2021), we further decouple the task-solving stage into reasoning with self-attention structures and extracting multilingual knowledge with feed-forward structures. Therefore, our hypothesized Multilingual Workflow (MWork) illustrated in Figure 2 outlines the three operational stages of LLMs in processing multilingual queries: Initially, LLMs *understand* queries by converting diverse linguistic features into a unified representation. In the *task-solving* phase, LLMs reason in English and incorporate multilingual knowledge to obtain factual content, using self-attention and feed-forward structures, respectively. Finally, models *generate* responses in the original language as the original query.

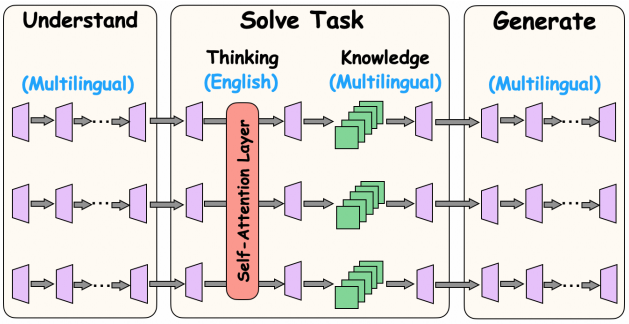


Figure 2: Our hypothesized multilingual workflow, MWork, converts multilingual queries to English for reasoning in English and generates responses in the original language, demonstrating a layered processing approach.

To verify the proposed MWork, we could extract language-specific parameters, selectively deactivate them within different structures, and observe their corresponding effects, thereby assessing the functionality of corresponding structures and validating our hypothesis. To identify the parameters to be activated, we develop a novel approach called Parallel Language-specific Neuron Detection (PLND). Unlike existing methods that rely on fine-tuning(Frankle and Carbin, 2018; Zhang et al., 2023b), labeled data (Tang et al., 2024; Liu et al., 2024), or parallel corpora (Libovický et al., 2020; Tanti et al., 2021; Zhang et al., 2024) to detect activated parameters, PLND measures the significance of individual neurons with respect to the input in both attention and feed-forward structures without any labeled data or parameter adjustments. Using PLND, we identify language-specific neurons by inputting a free text corpus of that language and isolating consistently activated neurons. We find that by deactivating language-specific neurons which account for only 0.13% of all neurons, LLMs' performance on a multilingual summarization task could drop by 99%.

We then extensively verify the hypothesized MWork framework using the proposed PLND method. Employing various benchmark tasks, including XQuAD (Artetxe et al., 2020) for understanding,

MGSM (Shi et al., 2022) for reasoning, X-CSQA (Lin et al., 2021) for knowledge extraction, and XLSum for generation (Hasan et al., 2021), we selectively deactivate language-specific neurons in each component and verify the functionality of the component by observing a significant decline in performance on the corresponding task. For example, when deactivating the language-specific neurons in the understanding layer, the performance on the multilingual understanding task XQuAD remains stable in English, while experiencing a decrease of 14% in non-English languages. Other tasks exhibit similar pattern when deactivating corresponding neurons. More importantly, with the verified `MWork` framework, enhancing the multilingual capabilities of LLMs can thus be achieved through the fine-tuning of language-specific neurons for certain capabilities. With a remarkable reduction in the training corpus size to a mere few hundred documents, this fine-tuning procedure enhances the multilingual capabilities of LLMs for both high-resource and low-resource languages by an average of 3.6% and 2.3% across all tasks, respectively. Notably, even without an English training corpus, there is a noticeable improvement in English performance, as the enhancement of language-specific neurons yields greater accuracy in enhancing specific languages, while simultaneously ensuring a clear division of parameters among different languages. In summary, the verified `MWork` reveals how LLMs handle multilingual tasks and offers an effective approach for conducting language-specific enhancements without compromising performance in other languages.

2 Parallel Language-specific Neuron Detection (`PLND`)

To verify the hypothesized workflow, we propose `PLND` that effectively detects language-specific neurons without relying on any labeled data. In essence, `PLND` identifies neurons crucial for handling individual documents, with language-specific neurons being those that consistently show high importance when processing documents in a particular language.

2.1 Sequential Neuron Detection

We define a neuron as a single row or column of a parameter matrix of a language model. To identify neurons responsible for a specific language, it is crucial to discern the significance of a neuron with respect to the inference of a given input. Specifically, when processing the input c in the model, we denote the hidden embedding before the i-th layer in Transformer (Vaswani et al., 2017) as h_i, and the hidden embedding after the i-th layer as $h_{i+1} = T_i(h_i)$, where T_i represents the parameters of the i-th layer. For a specific neuron within the i-th layer, denoted as $N^{(i)}$, either located in the attention or feed-forward network, we quantify its importance in processing the input c by measuring the difference in the hidden embedding after the i-th layer, i.e., h_{i+1}, when $N^{(i)}$ is activated or deactivated. Formally, the impact of neuron $N^{(i)}$ for input c is defined as

$$\text{Imp}(N^{(i)}|c) = \|T_i \backslash N^{(i)}(h_i) - T_i(h_i)\|_2, \tag{1}$$

where $T_i \backslash N^{(i)}(\cdot)$ denotes deactivating $N^{(i)}$ in T_i, i.e., setting all parameters of the neuron $N^{(i)}$ to zero. With a set of n corpus in a specific language, denoted as $\mathcal{C} = \{c_1, \cdots, c_l, \cdots, c_n\}$, we calculate the importance of each neuron in each layer to each corpus. Furthermore, we can obtain language-specific neurons that are important to all corpus in that language, i.e.,

$$\{N^{(i)} \mid \text{Imp}(N^{(i)}|c_l) \geq \epsilon, \forall c_l \in \mathcal{C}\}, \tag{2}$$

where ϵ is the pre-defined threshold.

2.2 Parallel Neuron Detection

The sequential neuron detection requires traversal of all neurons and inputs sequentially and thus is time-consuming. To address this, we further propose a parallel algorithm for accelerating the process.

Feed-Forward Network (FFN) In the latest open-source models, when processing input c, the feed-forward network in a certain layer is defined as

$$\text{FFN}(x) = \Big(\text{SiLU}\big(W_{gate}(x)\big) \cdot W_{up}(x)\Big) W_{down}, \tag{3}$$

where $x \in \mathbb{R}^{l \times d_{model}}$ is the embedding fed into the FFN, $W_{gate}, W_{up} \in \mathbb{R}^{d_{model} \times d_{inter}}$[2], $W_{down} \in \mathbb{R}^{d_{inter} \times d_{model}}$. The calculation of the importance of the k-th neuron in W_{up}, when processing the input c, as presented in Equation 1, can be equivalently transformed to

$$\text{Imp}(W_{up}[:,k]|c) = \|\hat{\text{FFN}}(x) - \text{FFN}(x)\|_2 = \Big\| \big(h_{\text{ffn}} \cdot \texttt{Mask}[k]\big) W_{down}(x) \Big\|_2, \tag{4}$$

where $h_{\text{ffn}} \in d_{inter}$ represents the embedding before W_{down}, and $\texttt{Mask}[k] \in d_{inter}$ is a vector with the k-th element equal to 1 and the rest equal to 0. To calculate $\text{Imp}(W_{up}[:,k]|c)$ for $k \in d_{inter}$ parallelly, we introduce a diagonal mask matrix of size (d_{inter}, d_{inter}), denoted as $\texttt{Mask}$. Therefore,

$$\text{Imp}(W_{up}|c) = \|(h_{\text{ffn}} \cdot \texttt{Mask}) W_{down}(x)\|_2. \tag{5}$$

Furthermore, we observe that deactivating the k-th neuron of W_{down} is equivalent to deactivating the k-th neuron in W_{up}, as they both result in $h_{\text{ffn}}[k] = 0$. Hence, we can also derive $\text{Imp}(W_{down}|c)$ by employing Equation (5).

Self-Attention Network When processing input c, the self-attention network in a certain layer is

$$\text{Attention}(x) = \text{Softmax}\big(\frac{W_Q(x)W_K^T(x)}{\sqrt{d}}\big) W_V(x), \tag{6}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d_{model} \times d_{mid}}$.[3] Since $W_V(x)$ is not in the non-linear softmax calculation, we can calculate $\text{Imp}(W_V|c)$ by applying Equation (5). For W_Q, we obtain $\text{Imp}(W_Q[:,k]|c)$ by deactivating its k-th neuron, specifically, $\hat{W}_Q \leftarrow W_Q[:,k] = 0$. Firstly, we calculate the difference in attention weight before and after deactivation, prior to scaling and softmax,

$$\Delta_k(x) = W_Q(x)W_K^T(x) - \hat{W}_Q(x)W_K^T(x) = W_Q(x)[:,k]W_K(x)[k,:] \in \mathbb{R}^{l \times l}. \tag{7}$$

Next, as the changes in attention exhibit a positive correlation with the changes in the output of this layer, the importance of $W_Q[:,k]$ in processing c, as defined in Equation 1, can be approximated as

$$\begin{aligned} \text{Imp}(W_Q[:,k]|c) &\approx \|\hat{\text{attention}}(x) - \text{attention}(x)\|_2 \\ &\approx \Big\| \text{softmax}\big(\frac{W_Q(x)W_K^T(x) - \Delta_k(x)}{\sqrt{d}}\big) - \text{softmax}\big(\frac{W_Q(x)W_K^T(x)}{\sqrt{d}}\big) \Big\|_2. \end{aligned} \tag{8}$$

This process can also be calculated in parallel, specifically,

$$\begin{aligned} \Delta(x) &= W_Q(x)W_K^T(x) - \hat{W}_Q(x)W_K^T(x) \\ &= W_Q(x).resize(l,1,d_{mid}) \times W_K(x).resize(1,l,d_{mid}) \in \mathbb{R}^{l \times l \times d_{mid}}. \end{aligned} \tag{9}$$

Therefore, the importance of W_Q in processing input c is calculated by

$$\text{Imp}(W_Q|c) \approx \Big\| \text{softmax}\big(\frac{W_Q(x)W_K^T(x) - \Delta(x)}{\sqrt{d}}\big) - \text{softmax}\big(\frac{W_Q(x)W_K^T(x)}{\sqrt{d}}\big) \Big\|_2. \tag{10}$$

Similarly, since W_K is symmetrical to W_Q, $\text{Imp}(W_K|c)$ can be calculated in the same way.

2.3 Detection of Language-Specific Neurons

We then apply PLND to selected languages and models to validate its effectiveness in detecting language-specific neurons and to further investigate the relationships between languages.

Experimental Setup. We test two open-source models that perform well on multilingual tasks, including *Vicuna-7b-v1.5*[4] (Chiang et al., 2023) and *Mistral-7b-Instruct-v0.2* (Jiang et al., 2023). For simplicity, we abbreviate them as Vicuna and Mistral hereafter to represent the two models respectively. We select the text summarization task with the XLSum (Hasan et al., 2021) dataset as the reference task to evaluate multilingual performance as it requires the model to comprehend the

[2] $W(\cdot)$ represents the linear matrix product of the input x and the parameter W, i.e., $W(x) := xW$.

[3] In some models like Vicuna and Mistral, $d_{model} = d_{mid}$, but we use different notations to avoid ambiguity.

[4] We do not directly utilize Llama2-chat as it does not follow multilingual instructions, consistently responding in English regardless of the language of the query.

Table 1: Multilingual performance on XLSum when deactivating language-specific neurons ("Lang-Spec") and an equivalent number of randomly selected neurons ("Random").

Model	Method	Fr	Zh	Es	Ru	Avg.
Vicuna	Original	14.2	61.1	10.4	20.8	26.6
	Deactivate Random	14.1	61.6	10.4	20.8	26.7
	Deactivate Lang-Spec	**0.83**	**0.00**	**0.24**	**0.42**	**0.37**
Mistral	Original	15.2	56.4	10.6	21.0	25.8
	Deactivate Random	15.4	55.9	10.2	21.2	25.7
	Deactivate Lang-Spec	**0.21**	**0.39**	**0.15**	**0.07**	**0.21**

input text and generate a coherent fragment. We adopt 4 high-resource languages including French (Fr), Chinese (Zh), Spanish (Es), and Russian (Ru), as their initial performance on those languages is already quite reasonable for observing the multilingual processing mechanism. Furthermore, we utilize OSCAR (Caswell et al., 2020) corpus which contains web crawling texts for each language to compile a language-specific corpus without task-specific considerations. More details are presented in Appendix B.

Existence of Language-Specific Neurons Using `PLND`, we feed a corpus in a specific language to LLMs and identify neurons that are consistently activated, which are responsible for processing queries in that language. To ascertain whether these neurons are genuinely language-specific, we assess the performance of LLMs in corresponding languages when these neurons are deactivated versus when the same number of randomly sampled neurons are deactivated.

Table 1 demonstrates the decline of multilingual capabilities when deactivating language-specific neurons. Although just deactivating around 0.13% neurons, LLMs lose their multilingual capabilities and fail to generate meaningful content. In contrast, deactivating the same number of randomly selected neurons does not yield any difference. Therefore, the detected neurons are language-specific and related to handling corresponding multilingual inputs.

2.4 Analysis of Language-Specific Neurons

We further investigate the degree of overlap among their language-specific neurons. Our findings reveal that in both Mistral and Vicuna, English shows limited overlap with other languages, indicating many language-specific neurons, while languages within the same family, such as Spanish, French, and English, demonstrate more overlap. More details are illustrated in Appendix C.

In addition, we examine two more types of multilingual LLMs, including BLOOMZ (Muennighoff et al., 2023), a *hyper-multilingual* LLM claiming to support 46 languages, and Chinese Llama (Cui et al., 2023), a *bilingual* LLM focusing on English and Chinese. We find that language-specific neurons in BLOOMZ follow patterns similar to Mistral and Vicuna. However, in Chinese LLama, Chinese dominates as the primary language for reasoning and knowledge extraction across all languages, with notably absent language-specific neurons. Details are shown in Appendix D.

Given the certain overlap ratio of language-specific neurons from other languages with those of English, as illustrated in the first column of Figure 5 and Figure 6, we conduct supplementary experiments to demonstrate that these neurons are not language-agnostic neurons crucial for general comprehension and logical reasoning (Liang et al., 2024; Tang et al., 2024). Instead, these overlapping neurons represent only a subset of language-specific neurons, while the language-agnostic neurons responsible for essential understanding and reasoning are those not identified as language-specific. Further elaboration and detailed results are presented in Appendix E.

3 Multilingual Workflow (`MWork`) of LLMs

3.1 `MWork`

By classifying the hidden representations of each layer in LLMs into English or non-English (as shown in Figure 1), we can observe the shift from non-English to English-centric, and back to non-English with the progression through the layers. This motivates us to hypothesize a three-stage multilingual workflow: *understanding* the original non-English queries and interpreting them in

English, *task-solving* in English, and *generating* back to the original language. Nevertheless, the presence of certain non-English tokens during the English-centric task-solving stage inspires us to further investigate this stage.

With the proposed PLND method, we extract language-specific neurons from attention and feed-forward structures when processing various multilingual queries. We plot the average number of activated language-specific neurons of Mistral when processing each query in Figure 3. Notably, the number of language-specific neurons decreases within the self-attention structure in the task-solving layer but remains consistent across the layers of the feed-forward structure. This decline implies a reliance on the English language for reasoning while extracting multilingual knowledge to support query processing, which is also consistent with (Geva et al., 2021)'s interpretation of the feed-forward structure as key-value memories for knowledge extraction. Therefore, we further decompose the task-solving layer into two parts: *reasoning* in English and *extracting knowledge* in a multilingual context.

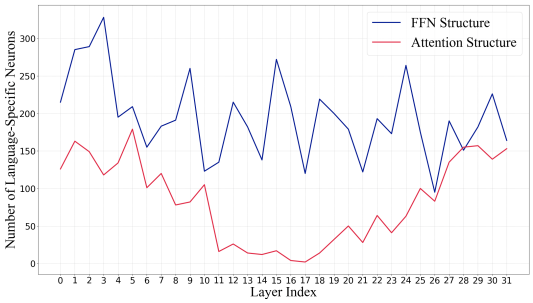


Figure 3: Number of language-specific neurons when processing multilingual queries.

Considering the above insights, we propose the MWork hypothesis for explaining LLM's multilingual workflow: LLMs first *understand* user input by unifying diverse linguistic features. They then engage in the *task-solving* phase, employing English for reasoning and leveraging multilingual knowledge through self-attention and feed-forward structures, respectively. Finally, the models *generate* responses aligned with the query's original language.

3.2 Verification Experiment Setup

To verify MWork, we selectively deactivate language-specific neurons from each component. Then its functionality can be verified if this deactivation results in minimal impact on English performance while exhibiting a notable decline in multilingual performance for the corresponding task.

Dataset To comprehensively understand how LLMs work with different abilities, we employ four kinds of tasks including MGSM (Shi et al., 2022) for reasoning task, XQuAD (Artetxe et al., 2020) for understanding task, X-CSQA (Lin et al., 2021) for knowledge question answering task, and XLSum (Hasan et al., 2021) for generation task. Detailed information regarding these datasets and the testing prompts can be found in Appendix F. We adopt 6 languages including English (En), German (De), French (Fr), Chinese (Zh), Spanish (Es), and Russian (Ru), as their initial performance on those languages is already quite reasonable for observing the multilingual processing mechanism. For XLSum, we randomly sample 500 data points from the whole test set for each language taking into consideration its long inference time, while for other tasks, we employ the entire test set. We evaluate the vanilla performance of Vicuna and Mistral on these datasets for later comparison as presented in Appendix G. For reasoning, understanding, and knowledge question answering tasks, we adopt accuracy as the metric. As for the generation tasks, we adopt ROUGE-L as the metric.

Deactivation Strategy We primarily consider two aspects when selecting the deactivation settings: (1) language-specific neurons versus randomly chosen neurons, and (2) the position of neurons, which encompasses four structures. Note that for a fair comparison, we ensure the numbers of deactivated neurons in all settings are the same. More detailed settings are explained from Section 3.3 to Section 3.6. For the concrete numbers of different layers, we tune hyperparameters by XQuAD in Chinese. Details are explained in Appendix H.

Notations Tables 2 to 5 present the results of deactivating certain neurons, where "Under" denotes the understanding layers, "S-ATTN" and "S-FFN" correspond to the self-attention and the feed-forward structures within the task-solving layers respectively, "Gen" refers to the generation layers. The term "Random" is used to describe deactivating randomly chosen neurons, whereas "Lang-Spec" refers to the deactivation of language-specific neurons. We also present the gap between the original performance (as shown in Table 11) and performance after deactivation (as shown in Table 14 to Table

Table 2: Results of the **understanding** task, where '✗' indicates that chosen neurons in the corresponding layer are deactivated, and '✓' signifies they are activated. Δ is defined as the difference between the reduction in performance in English, denoted as Δ_{Eng}, and the reduction in performance in non-English languages, denoted as $\Delta_{\text{n-Eng}}$.

Model	Deactivating Method					Performance				
	Under	S-ATTN	S-FFN	Gen	Neuron	Eng	n-Eng	Δ_{Eng}	$\Delta_{\text{n-Eng}}$	$\Delta\uparrow$
Vicuna	✗	✓	✓	✓	Random	57.8	53.9	+0.3	−0.1	+0.4
	✗	✗	✗	✗	Random	57.9	54.2	+0.4	+0.3	+0.1
	✓	✗	✗	✓	Lang-Spec	40.9	38.6	−15.9	−15.3	−0.6
	✓	✓	✓	✗	Lang-Spec	57.9	52.8	−0.4	−1.1	+0.7
	✗	✓	✓	✓	Lang-Spec	56.5	46.0	−0.5	−7.9	+7.4
Mistral	✗	✓	✓	✓	Random	58.1	55.5	+1.0	−0.2	+1.2
	✗	✗	✗	✗	Random	57.6	55.5	+0.5	−0.2	+0.7
	✓	✗	✗	✓	Lang-Spec	53.2	47.0	−3.9	−8.7	+4.8
	✓	✓	✓	✗	Lang-Spec	56.4	54.6	−0.7	−1.0	+0.3
	✗	✓	✓	✓	Lang-Spec	56.2	48.3	−0.9	−7.4	+6.5

17) for English (Δ_{Eng}) and averaged non-English languages ($\Delta_{\text{n-Eng}}$), respectively. A single metric Δ is then introduced as $\Delta_{\text{Eng}} - \Delta_{\text{n-Eng}}$, where a high value indicates such deactivation operation does not bring much impact to the English performance but lead to performance drop in non-English. Therefore, this provides a direct single indicator that the deactivated neurons are language-specific and hold a significant responsibility in executing the corresponding task.

3.3 Verify the Understanding Stage in MWork

Deactivating Method Table 2 shows the results of the understanding task following the deactivation of five distinct sets of neurons: (i) neurons randomly selected from the understanding layers; (ii) neurons randomly chosen across all layers; (iii) language-specific neurons within the task-solving layers; (iv) language-specific neurons in the generation layers; (v) language-specific neurons in the understanding layers. As mentioned above, in order to verify the functionality of the understanding layer (setting v), we compare it with deactivating other types of layers, specifically setting iii for the task-solving layer and setting iv for the generation layer. Full results are listed in Appendix I.

Findings We find that by deactivating randomly sampled neurons, no matter in the understanding layer or all layers, the performance of LLMs in both English and non-English languages is almost unaffected compared to other deactivating methods. Note that in some cases, deactivating randomly sampled neurons may even increase the performance because irrelevant neurons are removed, which also aligns with the finding from (Sharma et al., 2023). When assessing the differential impact on English and non-English language performance after the deactivation, specifically the difference calculated as $\Delta_{\text{Eng}} - \Delta_{\text{n-Eng}}$, it is evident that the deactivation of random neurons within the understanding layer amplifies this effect. This observation lends partial support to the hypothesized role of the understanding layer in language processing.

Furthermore, we find that deactivating language-specific neurons in the understanding layer influences the performance in English a little while significantly decreasing the performance in non-English languages. When deactivating language-specific neurons in the task-solving layer, both English and non-English languages are significantly reduced while deactivating language-specific neurons in the generation layer influences a little for both English and non-English languages. Therefore, we prove that the first several layers are responsible for understanding because deactivated neurons just disable LLMs on the NLU task in non-English languages. Furthermore, disabling language-specific neurons in the task-solving layer shows that LLMs rely on English, as performance drops across all languages.

3.4 Verify the Reasoning Structure in MWork

Deactivating Method Table 3 shows the result of the reasoning task, where we deactivate 6 sets of neurons. We adhere to the previous logic of selecting deactivation settings, with the exception that

Table 3: Results of the **reasoning** task. Disabling all language-specific neurons, except for those involved in self-attention structure within the task-solving layer, greatly reduces performance.

Model	Deactivating Method					Performance				
	Under	S-ATTN	S-FFN	Gen	Neuron	Eng	n-Eng	Δ_{Eng}	$\Delta_{\text{n-Eng}}$	$\Delta\uparrow$
Vicuna	✓	✗	✓	✓	Random	20.0	11.3	−0.4	−1.8	+1.4
	✓	✗	✗	✓	Random	18.4	12.2	−2.0	−1.0	−1.0
	✗	✗	✗	✗	Random	19.6	12.5	−0.8	−0.7	−0.1
	✓	✗	✗	✓	Lang-Spec	7.2	3.4	−13.2	−9.8	−3.4
	✗	✓	✓	✗	Lang-Spec	18.1	8.3	−2.3	−4.9	+2.6
	✗	✓	✗	✗	Lang-Spec	19.0	7.8	−1.4	−5.4	+4.0
Mistral	✓	✗	✓	✓	Random	40.8	23.4	−5.2	−2.9	−2.3
	✓	✗	✗	✓	Random	39.2	24.0	−6.8	−2.3	−4.5
	✗	✗	✗	✗	Random	45.2	26.8	−0.8	+0.5	−1.3
	✓	✗	✗	✓	Lang-Spec	38.2	18.4	−7.8	−7.9	+0.1
	✗	✓	✓	✗	Lang-Spec	44.0	18.1	−2.0	−8.2	+6.2
	✗	✓	✗	✗	Lang-Spec	46.2	18.3	+0.2	−8.0	+8.2

Table 4: Results of the **knowledge** question answering task. The highest performance reduction difference (Δ) is achieved by disabling all language-specific neurons in the feed-forward structure within the task-solving layer.

Model	Deactivating Method					Performance				
	Under	S-ATTN	S-FFN	Gen	Neuron	Eng	n-Eng	Δ_{Eng}	$\Delta_{\text{n-Eng}}$	$\Delta\uparrow$
Vicuna	✓	✓	✗	✓	Random	57.5	39.5	−0.3	+0.0	−0.3
	✓	✗	✗	✓	Random	56.0	38.7	−1.8	−0.8	−1.0
	✗	✗	✗	✗	Random	57.7	39.6	−0.1	+0.1	−0.2
	✓	✗	✓	✓	Lang-Spec	33.7	30.3	−24.1	−9.2	−14.9
	✓	✓	✗	✓	Lang-Spec	57.5	37.5	−0.3	−2.0	+1.7
Mistral	✓	✓	✗	✓	Random	61.0	37.0	−0.3	−0.5	+0.2
	✓	✗	✗	✓	Random	60.7	36.3	−0.6	−1.2	+0.6
	✗	✗	✗	✗	Random	61.8	37.4	+0.1	−0.1	+0.2
	✓	✗	✓	✓	Lang-Spec	51.2	28.9	−10.1	−8.6	−1.5
	✓	✓	✗	✓	Lang-Spec	61.2	35.1	−0.1	−2.4	+2.3

we do not conduct an independent experiment on deactivating neurons in the understanding layer, as its functionality has already been verified. Details are listed in Appendix I.

Findings We find that deactivating randomly sampled neurons in task-solving layers disables the capabilities of LLMs in reasoning to a greater extent than deactivating randomly sampled neurons in all layers, which verifies the function of the task-solving layer. Furthermore, comparing three deactivating language-specific neuron methods, we find that deactivating the task-solving layer decreases performance in both English and non-English. On the contrary, when we only deactivate language-specific neurons not in the task-solving layer, non-English is influenced more seriously than English. Moreover, eliminating interference from the feed-forward layer achieves better results, which verifies the function of attention structure in the task-solving layer.

3.5 Verify the Knowledge Extraction Structure in `MWork`

Deactivating Method Table 4 shows the result of the knowledge question answering task, where we deactivate 5 sets of neurons. Similarly, we exclude the deactivation of neurons in layers that have already been verified and instead concentrate on the self-attention structure and feed-forward structure in the task-solving layer. Details are listed in Appendix I.

Findings Likewise, targeted deactivation of language-specific neurons within the feed-forward structure of the task-solving layer predominantly affects non-English languages. This implies that

Table 5: Results of the **generation** task. The highest performance reduction difference (Δ) is achieved by disabling all language-specific neurons in the generation layer.

Model	Deactivating Method					Performance				
	Under	S-ATTN	S-FFN	Gen	Neuron	Eng	n-Eng	Δ_{Eng}	$\Delta_{\text{n-Eng}}$	$\Delta\uparrow$
Vicuna	✓	✓	✓	✗	Random	13.2	26.8	+0.1	+0.1	+0.0
	✗	✗	✗	✗	Random	13.0	26.7	−0.1	+0.0	−0.1
	✓	✓	✓	✗	Lang-Spec	13.1	25.7	+0.0	−1.1	+1.1
Mistral	✓	✓	✓	✗	Random	13.6	25.9	+0.1	+0.1	+0.0
	✗	✗	✗	✗	Random	13.6	25.7	+0.1	−0.2	+0.3
	✓	✓	✓	✗	Lang-Spec	13.8	24.3	+0.3	−1.5	+1.8

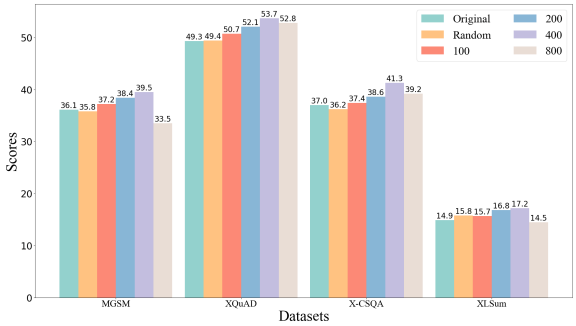


Figure 4: Enhancement results on high-resource languages, while the number is average among languages.

Table 6: Enhancement is achieved by fine-tuning Mistral-7b-v0.1 model utilizing 400 documents from each language correspondingly. The results are averaged across four tasks. Performance on English ("En") is obtained by averaging the results from four fine-tuned models.

Method	En	Vi	Th	Ar	Sw
Original	41.1	32.7	25.6	21.7	15.1
Random	40.8	32.7	25.2	21.2	15.1
Lang-Spec	**44.6**	**34.9**	**28.5**	**23.4**	**16.9**

processing multilingual queries necessitates accessing the multilingual information embedded within the relevant structures. However, disabling the self-attention structure compromises the ability to solve tasks across all languages.

3.6 Verify the Generation Structure in `MWork`

Deactivating Method Table 5 shows the result of the generation task, where we deactivate 3 sets of neurons. Since all previous layers have been verified, we solely deactivate neurons in the generation layer and compare them with randomly selected neurons. Details are listed in Appendix I.

Findings Similar to other tasks, the disabling of language-specific neurons within the generation layer diminishes their capacity to generate content in the respective languages. By selectively deactivating neurons that are not associated with English, we do not completely eliminate the models' multilingual generation abilities. However, as demonstrated in Table 1, the complete deactivation of all language-specific neurons results in the total loss of the LLMs' multilingual generation capabilities.

4 Multilingual Enhancement with `MWork`

We have verified `MWork` for explaining the multilingual working mechanism of LLMs in the above section via deactivating certain neurons. While opposite to employing deactivation, we can also enhance their multilingual ability, especially the understanding and generating ability, by fine-tuning these language-specific neurons. With language-specific neurons comprising only around 0.1% of all parameters, the need for training documents to improve multilingual capabilities can be significantly reduced to just a few hundred. Additionally, fine-tuning only the language-specific neurons for a particular language does not impact performance in other languages, allowing us to enhance specific languages while preserving performance in others.

`MWork` helps with enhancing multilingual ability by hundreds of documents. We employ *Mistral-7b-v0.1* for enhancement to eliminate the interference of instruction fine-tuning, and select causal language modeling as our training task. We create a dataset comprising $\{100, 200, 400, 800\}$

randomly selected documents for each language, extracted from the Wikipedia corpus (Foundation). Figure 4 shows the results of enhancement on high-resource languages (De, Fr, Zh, Es, Ru). The numbers represent the sizes of the training corpus when fine-tuning language-specific neurons, while "Random" represents the fine-tuning of an equivalent number of randomly chosen neurons using a corpus of 400. Our findings reveal that fine-tuning with a few hundred documents yields significant performance improvements on multilingual tasks: 3.4% on MGSM, 4.4% on XQuAD, 4.3% on X-CSQA, and 2.3% on XLSum. Moreover, English performance is enhanced by an average of 3.7% across all tasks. These results further confirm the effectiveness of `MWork` in interpreting structure functionality for LLM's multilingual query handling, offering precise and independent methods for multilingual enhancement. When fine-tuning with 800 documents, the performance deteriorates compared to using 400 documents. This drop can be attributed to the incorporation of additional knowledge, which disrupts the original knowledge distribution and leads to overfitting of the model to Wikipedia. This can be addressed by mixing data from more sources such as textbooks or websites.

In addition, we verify the effectiveness of such enhancement method on low-resource languages, given that low-resource performance is relatively low with the original model. We select four languages including Vietnamese (Vi), Thai (Th), Arabic (Ar), and Swahili (Sw), covering languages with both latin and non-latin scripts and having corresponding testing set in our considered benchmarks. The model was then evaluated on four benchmarks, and the result shown in Table 6 is the average scores among tasks. It is evident that the fine-tuning method using language-specific neurons enhances the model's multilingual performance in low-resource languages by an average of 2.2%. Notably, the improvement of 3.5% in English performance is observed even without an English training corpus, indicating the effectiveness of the distinct language responsibilities assigned to neurons.

5 Related Work

In the era of LLMs, numerous studies have been conducted to develop multilingual benchmarks (Zhang et al., 2023a), enhance multilingual performance without parameter adjustments through translation (Liang et al., 2023; Huang et al., 2023), aligning representations (Nguyen et al., 2023a; Salesky et al., 2023), prompting (Li et al., 2023b; Tanwar et al., 2023). Furthermore, certain works focus on improving multilingual abilities for a single task via cross-lingual transfer (Kim et al., 2017; Lin et al., 2019; Pfeiffer et al., 2020; Zhao et al., 2024b), while others aim to enhance multilingual proficiency by continuous training in one language to obtain mono-lingual LLMs (Cui et al., 2023), or in multiple domain languages to obtain domain-lingual LLMs (Nguyen et al., 2023b). Additionally, some works achieve multilingual LLMs by training from scratch (Muennighoff et al., 2023). However, these studies are limited to specific task types or require substantial training corpora due to a lack of comprehensive understanding of the multilingual mechanisms of LLMs.

Conventional interpretability research investigates the significance of input features with their corresponding outputs (Vig, 2019; Hewitt and Liang, 2019; Qiu et al., 2020). In the era of LLMs, one brunch of work includes efforts to understand knowledge storage, with (Geva et al., 2021) initiating the study of the feed-forward layer as a knowledge base. Subsequent work has furthered this by altering neuron values (Dai et al., 2022), mapping embeddings to words (Geva et al., 2022), modifying inputs to recover embeddings (Meng et al., 2022), and analyzing attention heads (Li et al., 2023a). Another line of research centers on the self-attention layer, examining its connection to reasoning capability (Hou et al., 2023; Stolfo et al., 2023; Friedman et al., 2023) by contrasting the reasoning tree based on attention weights.

6 Conclusion

In this work, we examine how LLMs handle multilingualism. The proposed multilingual workflow (`MWork`) suggests that LLMs initially understand queries by converting multilingual inputs into English, reason in English in intermediate layers while incorporating multilingual knowledge, and generate responses aligned with the original language in the final layers. The validity of `MWork` is verified using Parallel Language-specific Neuron Detection (`PLND`), which identifies activated neurons for different languages without labeled data. By detecting language-specific neurons and fine-tuning them with a small training corpus, `MWork` enhances multilingual abilities in specific languages without compromising others, resulting in significant improvements across tasks.

Acknowledgement

This work was substantially supported by DAMO Academy through DAMO Academy Research Intern Program. This research is partially supported by the National Research Foundation Singapore under the AI Singapore Programme (AISG Award No: AISG2-TC-2023-010-SGIL) and the Singapore Ministry of Education Academic Research Fund Tier 1 (Award No: T1 251RES2207).

References

Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2020. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4623–4637.

Isaac Caswell, Theresa Breiner, Daan van Esch, and Ankur Bapna. 2020. Language id in the wild: Unexpected challenges on the path to a thousand-language web text corpus. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6588–6608.

Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. 2023. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality.

Yiming Cui, Ziqing Yang, and Xin Yao. 2023. Efficient and effective text encoding for chinese llama and alpaca. *arXiv preprint arXiv:2304.08177*.

Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8493–8502.

Wikimedia Foundation. Wikimedia downloads.

Jonathan Frankle and Michael Carbin. 2018. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations*.

Dan Friedman, Andrew Lampinen, Lucas Dixon, Danqi Chen, and Asma Ghandeharioun. 2023. Interpretability illusions in the generalization of simplified models.

Mor Geva, Avi Caciularu, Kevin Wang, and Yoav Goldberg. 2022. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 30–45.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495.

Tahmid Hasan, Abhik Bhattacharjee, Md Saiful Islam, Kazi Mubasshir, Yuan-Fang Li, Yong-Bin Kang, M Sohel Rahman, and Rifat Shahriyar. 2021. Xl-sum: Large-scale multilingual abstractive summarization for 44 languages. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 4693–4703.

John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. *arXiv preprint arXiv:1909.03368*.

Yifan Hou, Jiaoda Li, Yu Fei, Alessandro Stolfo, Wangchunshu Zhou, Guangtao Zeng, Antoine Bosselut, and Mrinmaya Sachan. 2023. Towards a mechanistic interpretation of multi-step reasoning capabilities of language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4902–4919, Singapore. Association for Computational Linguistics.

Haoyang Huang, Tianyi Tang, Dongdong Zhang, Xin Zhao, Ting Song, Yan Xia, and Furu Wei. 2023. Not all languages are created equal in LLMs: Improving multilingual capability by cross-lingual-thought prompting. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 12365–12394, Singapore. Association for Computational Linguistics.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7b. *arXiv preprint arXiv:2310.06825*.

Joo-Kyung Kim, Young-Bum Kim, Ruhi Sarikaya, and Eric Fosler-Lussier. 2017. Cross-lingual transfer learning for pos tagging without cross-lingual resources. In *Proceedings of the 2017 conference on empirical methods in natural language processing*, pages 2832–2838.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023a. Inference-time intervention: Eliciting truthful answers from a language model. *arXiv preprint arXiv:2306.03341*.

Shuang Li, Xuming Hu, Aiwei Liu, Yawen Yang, Fukun Ma, Philip S Yu, and Lijie Wen. 2023b. Enhancing cross-lingual natural language inference by soft prompting with multilingual verbalizer. *arXiv preprint arXiv:2305.12761*.

Yaobo Liang, Quanzhi Zhu, Junhe Zhao, and Nan Duan. 2023. Machine-created universal language for cross-lingual transfer. *arXiv preprint arXiv:2305.13071*.

Yunlong Liang, Fandong Meng, Songming Zhang, Yufeng Chen, Jinan Xu, Jie Zhou, et al. 2024. Multilingual knowledge editing with language-agnostic factual neurons. *arXiv preprint arXiv:2406.16416*.

Jindřich Libovickỳ, Rudolf Rosa, and Alexander Fraser. 2020. On the language neutrality of pre-trained multilingual representations. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1663–1674.

Bill Yuchen Lin, Seyeon Lee, Xiaoyang Qiao, and Xiang Ren. 2021. Common sense beyond english: Evaluating and improving multilingual language models for commonsense reasoning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1274–1287.

Yu-Hsiang Lin, Chian-Yu Chen, Jean Lee, Zirui Li, Yuyan Zhang, Mengzhou Xia, Shruti Rijhwani, Junxian He, Zhisong Zhang, Xuezhe Ma, et al. 2019. Choosing transfer languages for cross-lingual learning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, volume 57.

Weize Liu, Yinlong Xu, Hongxia Xu, Jintai Chen, Xuming Hu, and Jian Wu. 2024. Unraveling babel: Exploring multilingual activation patterns within large language models. *arXiv preprint arXiv:2402.16367*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in gpt. *Advances in Neural Information Processing Systems*, 35:17359–17372.

Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao, M Saiful Bari, Sheng Shen, Zheng Xin Yong, Hailey Schoelkopf, et al. 2023. Crosslingual generalization through multitask finetuning. In *The 61st Annual Meeting Of The Association For Computational Linguistics*.

Hoang H Nguyen, Chenwei Zhang, Tao Zhang, Eugene Rohrbaugh, and Philip S Yu. 2023a. Enhancing cross-lingual transfer via phonemic transcription integration. *arXiv preprint arXiv:2307.04361*.

Xuan-Phi Nguyen, Wenxuan Zhang, Xin Li, Mahani Aljunied, Qingyu Tan, Liying Cheng, Guanzheng Chen, Yue Deng, Sen Yang, Chaoqun Liu, et al. 2023b. Seallms–large language models for southeast asia. *arXiv preprint arXiv:2312.00738*.

OpenAI. 2023. Gpt-4 technical report.

Jonas Pfeiffer, Ivan Vulić, Iryna Gurevych, and Sebastian Ruder. 2020. Mad-x: An adapter-based framework for multi-task cross-lingual transfer. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7654–7673.

Xipeng Qiu, Tianxiang Sun, Yige Xu, Yunfan Shao, Ning Dai, and Xuanjing Huang. 2020. Pre-trained models for natural language processing: A survey. *Science China Technological Sciences*, 63(10):1872–1897.

Elizabeth Salesky, Neha Verma, Philipp Koehn, and Matt Post. 2023. Pixel representations for multilingual translation and data-efficient cross-lingual transfer. *arXiv preprint arXiv:2305.14280*.

Pratyusha Sharma, Jordan T Ash, and Dipendra Misra. 2023. The truth is in there: Improving reasoning in language models with layer-selective rank reduction. *arXiv preprint arXiv:2312.13558*.

Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, et al. 2022. Language models are multilingual chain-of-thought reasoners. In *The Eleventh International Conference on Learning Representations*.

Alessandro Stolfo, Yonatan Belinkov, and Mrinmaya Sachan. 2023. A mechanistic interpretation of arithmetic reasoning in language models using causal mediation analysis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7035–7052, Singapore. Association for Computational Linguistics.

Tianyi Tang, Wenyang Luo, Haoyang Huang, Dongdong Zhang, Xiaolei Wang, Xin Zhao, Furu Wei, and Ji-Rong Wen. 2024. Language-specific neurons: The key to multilingual capabilities in large language models. *arXiv preprint arXiv:2402.16438*.

Marc Tanti, Lonneke van der Plas, Claudia Borg, and Albert Gatt. 2021. On the language-specificity of multilingual bert and the impact of fine-tuning. In *Proceedings of the Fourth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pages 214–227.

Eshaan Tanwar, Manish Borthakur, Subhabrata Dutta, and Tanmoy Chakraborty. 2023. Multilingual llms are better cross-lingual in-context learners with alignment. *arXiv preprint arXiv:2305.05940*.

Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Jesse Vig. 2019. A multiscale visualization of attention in the transformer model.

Wenxuan Zhang, Sharifah Mahani Aljunied, Chang Gao, Yew Ken Chia, and Lidong Bing. 2023a. M3exam: A multilingual, multimodal, multilevel benchmark for examining large language models. *CoRR*, abs/2306.05179.

Zhihao Zhang, Jun Zhao, Qi Zhang, Tao Gui, and Xuanjing Huang. 2024. Unveiling linguistic regions in large language models. *arXiv preprint arXiv:2402.14700*.

Zhong Zhang, Bang Liu, and Junming Shao. 2023b. Fine-tuning happens in tiny subspaces: Exploring intrinsic task-specific subspaces of pre-trained language models. In *The 61st Annual Meeting Of The Association For Computational Linguistics*.

Jun Zhao, Zhihao Zhang, Luhui Gao, Qi Zhang, Tao Gui, and Xuanjing Huang. 2024a. Llama beyond english: An empirical study on language capability transfer.

Yiran Zhao, Wenxuan Zhang, Huiming Wang, Kenji Kawaguchi, and Lidong Bing. 2024b. Adamergex: Cross-lingual transfer with large language models via adaptive adapter merging. *arXiv preprint arXiv:2402.18913*.

A English and Non-English Tokens

We employ `cld3` package to detect the language of each token in the embeddings of each layer, which is a language detection library based on the Compact Language Detector 3 model developed by Google. Furthermore, if the detection result is reliable, i.e., cld3.get_language(token).is_reliable $== True$, we adopt the detection results, otherwise the token is categorized as a non-word.

B Multilingual Corpus

Note that our selection criterion for the number of documents is based on achieving substantial coverage of each language's vocabulary, ensuring that the selected contexts provide a representative sample of the language, as shown in Table 7.

Table 7: Corpus details across languages are tailored to encompass the majority of each language's vocabulary, where "corpus size" indicates the number of contexts selected, "corpus vocab" represents the vocabulary coverage within the selected contexts, "vocab size" refers to the number of vocabularies of that language.

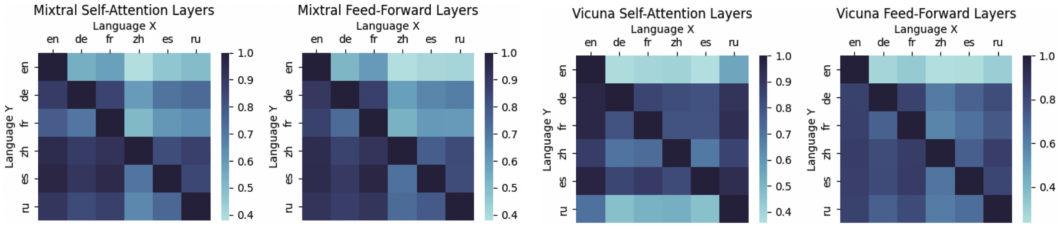
Language	En	De	Fr	Zh	Es	Ru
Corpus Size	180k	30k	50k	20k	20k	20k
Corpus Vocab	249k	154k	134k	198k	90k	144k
Vocab Size	273k	148k	135k	329k	93k	150k

C Interrelation of Language-Specific Neurons Across Languages

Using neurons identified by `PLND`, we investigate the relationships between languages via the degree of overlap among their language-specific neurons, defined as

$$\text{overlap}(x, y) = \frac{|\mathcal{N}_x \cap \mathcal{N}_y|}{|\mathcal{N}_y|}, \tag{11}$$

where $\mathcal{N}_{language}$ represents the set of detected language-specific neurons. Figure 5 shows the neuron overlapping ratio $\text{overlap}(x, y)$ of any two languages in different structures of two models.



(a) Mistral-7B-Instruct-v0.2.

(b) Vicuna-7b-v1.5.

Figure 5: Overlapping ratio of language-specific neurons in self-attention and feed-forward structures.

We can observe that in both Mistral and Vicuna, the intersection with English from other languages is relatively limited (i.e., the first row of each figure), suggesting that English possesses a predominant number of language-specific neurons. Additionally, there is a pronounced tendency for languages belonging to the same family to demonstrate a higher degree of overlap with each other, such as Spanish, French, and English.

D Analysis on Different Multilingual LLMs

We further examine two more types of multilingual LLMs, including BLOOMZ (Muennighoff et al., 2023), a *hyper-multilingual* LLM claiming to support 46 languages, and Chinese Llama (Cui et al., 2023), a *bilingual* LLM focusing on English and Chinese.

Hyper-Multilingual LLMs Figure 6 illustrates the degree of neuron overlap among languages within both the self-attention and feed-forward structures of BLOOMZ. In contrast to the findings shown in Figure 5, there is a marked reduction in overlap, indicating that individual languages maintain a higher degree of independence and do not extensively share neurons with one another.

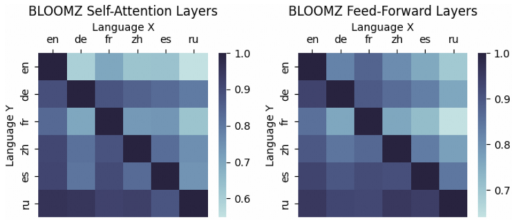


Figure 6: Overlapping ratio of language-specific neurons in BLOOMZ

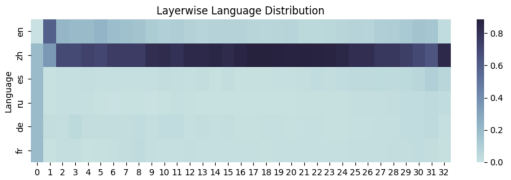


Figure 7: Ratio of languages among layers in Chinese Llama given non-English instructions.

Bilingual LLMs We employ Chinese Llama (Cui et al., 2023), which extends existing vocabulary and incorporate secondary pre-training using Chinese data and fine-tune the model with Chinese instruction datasets. However, this intensive training can lead to a degradation in performance for languages other than Chinese. As depicted in Figure 7, Chinese predominates as the primary language for reasoning processing and knowledge extraction across all languages. Consequently, the absence of language-specific neurons results in the transformation of it into a Chinese-centric LLM.

E Language-Agnostic Neurons

We initially implement a radical deactivation approach, wherein we specifically deactivate overlapping elements between each language and English. These elements precisely correspond to the intersecting neurons in the first column of Figure 5. Presented below are the comprehensive findings pertaining to Mistral. Our evaluation is centered around the reasoning task, which is recognized as the most indicative and challenging assessment for the model. We compare under the optimal "deactivating" method, which involves deactivating all language-specific neurons except those in S-ATTN.

Table 8: Performance of deactivating language-specific neurons without overlapped between English.

Language	Eng	non-Eng	Δ_{Eng}	$\Delta_{\text{non-Eng}}$	$\Delta\uparrow$
All language-specific neurons	46.2	18.3	$+0.2$	-8.0	$+8.2$
LSN without overlapped between English	45.8	20.2	-0.2	-6.1	$+5.9$

As evident by Table 8, the performance of English remains stable, contrasting sharply with the significant decline in the performance of multilingual. Removing overlapped neurons, as opposed to deactivating all language-specific neurons, leads to a less pronounced drop, yet the impact remains noteworthy. This demonstrates that overlapped neurons are not language-agnostic; they are not utilized for general comprehension and logical reasoning. Otherwise, the fundamental reasoning capacity and performance in multilingual contexts would remain unaffected. In addition, we retained the language-specific neurons that overlapped in all languages, meaning that we removed them from the language-specific neurons to be deactivated. Detailed results follow.

Table 9: Performance of deactivating language-specific neurons without all languages overlapped.

Language	Eng	non-Eng	Δ_{Eng}	$\Delta_{\text{non-Eng}}$	$\Delta\uparrow$
All language-specific neurons	46.2	18.3	$+0.2$	-8.0	$+8.2$
LSN without all languages overlapped	45.6	18.7	-0.4	-7.6	$+7.2$

The neurons that overlap across all languages only account for 0.02% of the total number of neurons. From the results in Table 9, we can see that the performance is almost the same as deactivating all language-specific neurons. This further proves that these neurons are not language-agnostic neurons, but only a subset of language-specific neurons.

Table 10: Zero-shot prompts for each dataset.

Task	Zero-Shot Prompt
MGSM	Let's think step by step. Question: {question}
XQuAD	{context} Question: {question}
XLSum	Summarize the context in one sentence. Title: {title} Context: {article}
X-CSQA	Question: {question}

Table 11: Assessing the baseline performance of Vicuna and Mistral across four representative multilingual tasks in selected languages, where Avg. is calculated among non-English languages.

Model	Task	En	De	Fr	Zh	Es	Ru	Avg.
Vicuna	XQuAD	57.5	50.3	–	55.7	55.7	–	53.9
	MGSM	20.4	14.8	14.8	12.8	13.2	10.0	13.1
	X-CSQA	57.8	43.8	40.1	43.2	44.3	26.0	39.5
	XLSum	13.1	–	14.2	61.1	10.4	20.8	26.6
Mistral	XQuAD	57.1	48.5	–	64.3	54.1	–	55.6
	MGSM	46.0	21.2	26.0	31.6	31.2	21.6	26.3
	X-CSQA	61.7	40.0	40.4	47.1	45.7	14.1	37.5
	XLSum	13.5	–	15.2	56.4	10.6	21.0	25.8

F Prompts

Table 10 shows the zero-shot prompts for each dataset. Note that when conducting tests in other languages, prompts are translated into the respective languages.

G Original Performance

Table 11 shows the original performance of Vanilla and Mistral on four tasks.

H Hyper-parameters

We adopt the performance on XQuAD in Chinese as the validation set to all languages and all tasks. Specifically, Table 12 shows the result on Vicuna when deactivating language-specific neurons in the understanding layer ($D_{\mathcal{U}}$) and generation layer ($D_{\mathcal{G}}$), where N_1 is the number of understanding layers and N_2 is the number of generation layer. We find that when setting $N_1 = 8$ and $N_2 = 2$, performance in English is influenced the least while performance in Chinese decreases the most. As for Mistral, the number is $N_1 = 6$ and $N_2 = 3$.

Table 12: XQuAD with Chinese on Vicuna.

Method	$D_{\mathcal{U}}$ N_1	$D_{\mathcal{U}}$ ACC	$D_{\mathcal{G}}$ N_2	$D_{\mathcal{G}}$ ACC
En-Vanilla		57.5		
Zh-Vanilla		55.5		
En-Deact	**8**	**57.7** (↑ 0.2)	4	54.7 (↓ 2.8)
Zh-D-Deact		**44.9** (↓ 10.6)		54.6 (↓ 0.9)
En-Deact	6	58.6 (↑ 1.1)	3	57.7 (↑ 0.2)
Zh-Deact		55.1 (↓ 0.4)		54.5 (↓ 1.0)
En-Deact	4	57.3 (↓ 0.2)	**2**	**58.4** (↑ 0.9)
Zh-Deact		53.9 (↓ 1.6)		**54.1** (↓ 1.4)

Table 13: XQuAD with Chinese on Mistral.

Method	$D_{\mathcal{U}}$ N_1	$D_{\mathcal{U}}$ ACC	$D_{\mathcal{G}}$ N_2	$D_{\mathcal{G}}$ ACC
En-Vanilla		57.1		
Zh-Vanilla		64.3		
En-Deact	8	53.3 (↓ 3.8)	4	55.8 (↓ 1.3)
Zh-Deact		52.6 (↓ 11.7)		62.9 (↓ 1.4)
En-Deact	**6**	**56.8** (↓ 0.3)	**3**	**56.3** (↓ 0.8)
Zh-Deact		**54.9** (↓ 9.4)		**62.7** (↓ 1.6)
En-Deact	4	57.6 (↑ 0.5)	2	55.7 (↓ 1.4)
Zh-Deact		61.8 (↓ 2.5)		63.8 (↓ 0.5)

I Detailed Experiment Results

I.1 Detailed Experiment Settings

Reasoning Task Deactivation methods: (i) randomly sampled neurons in the attention structure of task-solving layer. (ii) randomly sampled neurons in the task-solving layer. (iii) randomly sampled neurons in all layers. (iv) language-specific neurons in the task-solving layer. (v) language-specific neurons in the understanding layer and generation layer. (vi) language-specific neurons not in the attention structure of task-solving layers.

Knowledge Question Answering Task Deactivation methods: (i) randomly sampled neurons in the feed-forward structure of task-solving layers. (ii) randomly sampled neurons in the task-solving layer. (iii) randomly sampled neurons in all layers. (iv) language-specific neurons in the attention structure of task-solving layers. (v) language-specific neurons in the feed-forward structure of task-solving layers.

Generation Task Deactivation methods: (i) randomly sampled neurons in the generating layers. (ii) randomly sampled neurons in all layers. (iv) language-specific neurons in the generating layers.

I.2 Detailed Result

Due to the limited space, we employ a more concise notation. We denote deactivating neurons in the self-attention layer of the i-th layer as $D_i^{(A)}$, while deactivating neurons in the feed-forward layer of the i-th layer is denoted as $D_i^{(F)}$. We denote $\mathcal{U} = \{1, \cdots, N_1\}$ as the set of layers that take charge of understanding as shown in Figure 2. Similarly, we denote $\mathcal{S} = \{N_1 + 1, \cdots, N_2\}$ as the set of layers that take charge of task solving and $\mathcal{G} = \{N_2 + 1, \cdots, 32\}$ as the set of layers that take charge of generation[5]. Furthermore, $D_{\mathcal{U}}^{(A)}$ represents deactivating neurons in self-attention layers of $\mathcal{U}$. Similarly, we introduce $D_{\mathcal{U}}^{(F)}$, $D_{\mathcal{S}}^{(A)}$, $D_{\mathcal{S}}^{(F)}$, $D_{\mathcal{G}}^{(A)}$ and $D_{\mathcal{G}}^{(A)}$.

Table 14: Understanding task.

	Method	German				Chinese				Spanish			
		En-D	De-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{De-D}}$	En-D	Zh-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Zh-D}}$	En-D	Es-D	$\Delta_{\text{Es-D}}$	$\Delta_{\text{Es-D}}$
Vicuna	$D_{\mathcal{U}}^{R}$	57.8	49.7	+0.3	−0.6	57.8	55.8	+0.3	+0.1	57.8	56.1	+0.3	+0.4
	D_{All}^{R}	57.9	50.8	+0.4	+0.5	57.9	55.8	+0.4	+0.1	57.9	55.9	+0.4	+0.2
	$D_{\mathcal{U}}$	55.7	40.7	−2.0	−9.6	57.7	44.9	+2.0	−10.8	56.1	52.4	−1.4	−3.2
	$D_{\mathcal{S}}$	48.3	41.7	−7.2	−8.6	45.0	45.4	−12.5	−10.3	29.5	28.6	−28.0	−27.1
	$D_{\mathcal{G}}$	57.5	50.1	0.0	−0.2	58.4	54.1	+0.9	−1.6	57.7	54.1	+0.2	−1.6
Mistral	$D_{\mathcal{U}}^{R}$	58.1	48.2	+1.0	−0.4	58.1	63.9	+1.0	−0.4	58.1	54.3	+1.0	+0.2
	D_{All}^{R}	57.6	48.3	+0.5	−0.3	57.6	63.6	+0.5	−0.7	57.6	54.5	+0.5	+0.4
	$D_{\mathcal{U}}$	56.5	42.4	−0.6	−6.2	56.8	54.9	−0.3	−9.4	55.4	47.5	−1.7	−6.6
	$D_{\mathcal{S}}$	54.3	43.2	−2.8	−5.4	54.9	52.9	−2.2	−11.4	50.3	44.9	−6.8	−9.2
	$D_{\mathcal{G}}$	56.7	47.9	−0.4	−0.7	56.3	62.7	−0.8	−1.6	56.2	53.2	−0.9	−0.8

[5]Vicuna-7b-v1.5 and Mistral-7b-v1.0 both have 32 layers.

Table 15: Reasoning task.

	Method	German				French				Chinese				Spanish				Russian			
		En-D	De-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{De-D}}$	En-D	Fr-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Fr-D}}$	En-D	Zh-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Zh-D}}$	En-D	Es-D	$\Delta_{\text{Es-D}}$	$\Delta_{\text{Es-D}}$	En-D	Ru-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Ru-D}}$
Vicuna	$D^R_{\mathcal{S}^{(A)}}$	20.0	12.4	−0.4	−2.4	20.0	13.6	−0.4	−1.2	20.0	13.2	−0.4	+0.4	20.0	12.4	−0.4	−0.8	20.0	4.8	−0.4	−5.2
	$D^R_{\mathcal{S}}$	18.4	12.4	−2.0	−2.4	18.4	14.0	−2.0	−0.8	18.4	14.4	−2.0	+1.6	18.4	15.2	−2.0	+2.0	18.4	4.8	−2.0	−5.2
	D^R_{All}	19.6	14.0	−0.8	−0.8	19.6	13.8	−0.8	−1.0	19.6	14.8	−0.8	+2.0	19.6	12.4	−0.8	−0.8	19.6	7.6	−0.8	−2.4
	$D_{\mathcal{S}}$	3.6	2.0	−16.8	−12.8	8.4	3.2	−12.0	−11.6	4.8	4.0	−15.6	−8.8	8.8	4.0	−11.6	−9.2	10.4	4.0	−10.0	−6.0
	$D_{\mathcal{U}\&\mathcal{G}}$	16.4	5.6	−4.0	−9.2	19.2	9.6	−1.2	−5.2	20.0	9.2	−0.4	−3.6	17.6	11.6	−2.8	−1.6	17.2	5.6	−3.2	−4.4
	$\bar{D}_{\mathcal{S}^{(A)}}$	16.8	4.4	−3.6	−10.4	19.6	8.8	−0.8	−4.4	21.6	9.6	+1.2	−3.2	19.6	10.4	−0.8	−2.8	17.2	5.6	−3.2	−4.4
Mistral	$D^R_{\mathcal{S}^{(A)}}$	40.8	18.0	−5.2	−3.2	40.8	25.6	−5.2	−0.4	40.8	24.0	−5.2	−7.6	40.8	29.2	−5.2	−2.0	40.8	20.4	−5.2	−1.2
	$D^R_{\mathcal{S}}$	39.2	20.0	−6.8	−1.2	39.2	25.2	−6.8	−0.8	39.2	25.6	−6.8	−6.0	39.2	29.6	−6.8	−1.6	39.2	19.6	−6.8	−2.0
	D^R_{All}	45.2	24.0	−0.8	+2.8	45.2	27.6	−0.8	+1.6	45.2	31.2	−0.8	−0.4	45.2	30.4	−0.8	−0.8	45.2	20.8	−0.8	−0.8
	$D_{\mathcal{S}}$	38.4	12.0	−7.6	−9.2	40.8	24.8	−5.2	−1.2	37.9	19.6	−8.1	−12.0	40.4	24.4	−5.6	−6.8	33.6	11.2	−12.4	−10.4
	$D_{\mathcal{U}\&\mathcal{G}}$	42.4	9.2	−3.6	−12.0	41.2	21.6	−4.8	−4.4	46.4	19.6	+0.4	−12.0	44.0	28.0	−2.0	−3.2	46.0	12.0	+0.0	−9.6
	$\bar{D}_{\mathcal{S}^{(A)}}$	43.6	9.6	−2.4	−11.6	44.8	19.2	−1.2	−6.8	46.4	18.8	+0.4	−12.8	47.6	27.6	+1.6	−3.6	48.4	16.4	+2.4	−5.2

Table 16: Knowledge Question Answering task.

	Method	German				French				Chinese				Spanish				Russian			
		En-D	De-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{De-D}}$	En-D	Fr-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Fr-D}}$	En-D	Zh-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Zh-D}}$	En-D	Es-D	$\Delta_{\text{Es-D}}$	$\Delta_{\text{Es-D}}$	En-D	Ru-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Ru-D}}$
Vicuna	$D^R_{\mathcal{S}^{(F)}}$	57.5	43.8	−0.3	+0.0	57.5	40.3	−0.3	+0.2	57.5	43.2	−0.3	+0.0	57.5	44.6	−0.3	+0.3	57.5	25.5	−0.3	−0.5
	$D^R_{\mathcal{S}}$	56.0	44.0	−1.8	+0.2	56.0	38.6	−1.8	−1.5	56.0	43.4	−1.8	+0.2	56.0	43.5	−1.8	−0.8	56.0	24.0	−1.8	−2.0
	D^R_{All}	57.7	43.6	−0.1	−0.2	57.7	40.5	−0.1	+0.4	57.7	43.2	−0.1	+0.0	57.7	44.5	−0.1	+0.2	57.7	26.0	−0.1	+0.0
	$D_{\mathcal{S}^{(A)}}$	34.8	43.4	−23.0	−0.4	32.6	31.1	−25.2	−12.7	32.6	28.9	−25.2	−14.3	20.4	25.0	−37.1	−19.3	48.3	22.9	−9.5	−3.1
	$D_{\mathcal{S}^{(F)}}$	57.8	41.5	+0.0	−2.5	57.2	37.8	−0.6	−6.0	56.9	39.6	−0.9	−3.6	57.6	43.0	−0.2	−1.3	57.8	25.6	+0.0	−0.4
Mistral	$D^R_{\mathcal{S}^{(F)}}$	61.0	40.2	−0.7	+0.2	61.0	40.1	−0.7	−0.3	61.0	46.7	−0.7	−0.4	61.0	45.2	−0.7	−0.5	61.0	12.7	−0.7	−1.4
	$D^R_{\mathcal{S}}$	60.7	40.4	−1.0	+0.4	60.7	36.9	−1.0	−3.5	60.7	46.9	−1.0	−0.3	60.7	46.3	−1.0	+0.7	60.7	11.1	−1.0	−3.0
	D^R_{All}	61.8	40.1	+0.1	+0.1	61.8	40.7	+0.1	+0.3	61.8	47.2	+0.1	+0.1	61.8	44.7	+0.1	−1.0	61.8	14.1	+0.1	+0.0
	$D_{\mathcal{S}^{(A)}}$	50.4	32.3	−11.3	−7.7	55.3	27.4	−6.4	−13.0	54.7	42.4	−7.0	−4.7	44.5	34.1	−17.2	−11.6	51.1	8.3	−10.6	−5.8
	$D_{\mathcal{S}^{(F)}}$	61.5	38.1	−0.2	−1.9	61.2	38.1	−0.5	−2.3	61.3	43.5	−0.4	−3.6	61.0	43.9	−0.7	−1.8	60.8	11.8	−0.4	−2.3

Table 17: Generation task.

	Method	French				Chinese				Spanish				Russian			
		En-D	Fr-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Fr-D}}$	En-D	Zh-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Zh-D}}$	En-D	Es-D	$\Delta_{\text{Es-D}}$	$\Delta_{\text{Es-D}}$	En-D	Ru-D	$\Delta_{\text{En-D}}$	$\Delta_{\text{Ru-D}}$
Vicuna	$D^R_{\mathcal{G}}$	13.2	14.2	+0.1	+0.0	13.2	61.6	+0.1	+0.5	13.2	10.4	+0.1	+0.0	13.2	20.8	+0.1	+0.0
	D^R_{All}	13.0	14.1	−0.1	−0.1	13.0	61.6	−0.1	+0.5	13.0	10.4	−0.1	+0.0	13.0	20.8	−1.0	+0.0
	$D_{\mathcal{G}}$	13.0	13.8	−0.1	−0.4	13.1	59.5	+0.0	−1.6	13.0	9.1	−0.1	−1.3	13.1	20.3	+0.0	−0.5
Mistral	$D^R_{\mathcal{G}}$	13.6	15.2	+0.1	+0.0	13.6	56.7	+0.1	+0.3	13.6	10.3	+0.1	−0.3	13.6	21.2	+0.1	+0.2
	D^R_{All}	13.6	15.4	+0.1	+0.2	13.6	55.9	+0.1	−0.5	13.6	10.2	+0.1	−0.4	13.6	21.1	+0.1	+0.1
	$D_{\mathcal{G}}$	14.3	14.2	+0.8	−1.0	13.6	52.8	+0.1	−3.6	13.7	10.2	+0.2	−0.4	13.5	20.2	−0.1	−0.8

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: In the abstract and Section 1 (Introduction).

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: After Section 6 (Conclusion)

   Guidelines:

   - The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

Justification: The paper does not include theoretical results.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: In Section 2, 3 and 4.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Our code is available at https://github.com/DAMO-NLP-SG/multilingual_analysis. We will provide sufficient details to fully reproduce our results.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (https://nips.cc/public/guides/CodeSubmissionPolicy) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://nips.cc/public/guides/CodeSubmissionPolicy) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

   Justification: In Section 2, 3 and 4.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: There is no randomness in our experiments

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).

- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: In Section 2, 3 and 4.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code Of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: This paper adheres to the NeurIPS Code of Ethics.

   Guidelines:

   - The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [NA]

    Justification: There is no societal impact of the work performed.

    Guidelines:

    - The answer NA means that there is no societal impact of the work performed.
    - If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [NA]

    Justification: The paper poses no such risks

    Guidelines:

    - The answer NA means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: In Section 2, 3 and 4.

    Guidelines:

    - The answer NA means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [NA]

    Justification: The paper does not release new assets.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: The paper does not involve crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: The paper does not involve crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.